

## **Project Title**

# **OptiCrop – Smart Agricultural Production Optimization Engine**

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## Introduction

OptiCrop is an intelligent Machine Learning-based web application developed to recommend the most suitable crop for a given plot of land. The system helps farmers, agribusinesses, and agricultural researchers make faster, more accurate, and data-driven planting decisions by analyzing soil and climate readings and predicting the crop most likely to succeed.

The application leverages a curated agricultural dataset and multiple analytical techniques – including K-Means clustering to uncover soil/climate patterns and a Logistic Regression classifier for crop prediction – to identify the most suitable crop for a given input profile. The best-performing model is integrated into a Flask-based web application that enables users to enter soil and climate readings and receive an instant crop recommendation.

The system evaluates various factors such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, soil pH, rainfall, and seasonal patterns before generating a prediction.

To enhance the user experience, the application provides a modern and responsive interface where users can submit their soil and climate readings, view the recommended crop on the same page, and read a short description of why that crop suits their conditions. This feature makes the system more informative and user-friendly by helping farmers understand and act on the recommendation.

## Ideation Phase

### Brainstorm & Idea Prioritization Template

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
| Name          | K.Hemanth Kumar                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 4 Marks                                                      |

### Brainstorm & Idea Prioritization Template:

OptiCrop is a Machine Learning-powered web application designed to recommend the most suitable crop for cultivation based on soil nutrients and climate conditions. The system helps farmers, agribusinesses, and agricultural researchers make faster, more reliable, data-driven planting decisions by analyzing soil and environmental readings and determining which crop is most likely to give the best yield.

The application evaluates soil and climate patterns using clustering and classification algorithms and deploys the best-performing model through a Flask-based web interface for real-time recommendations.

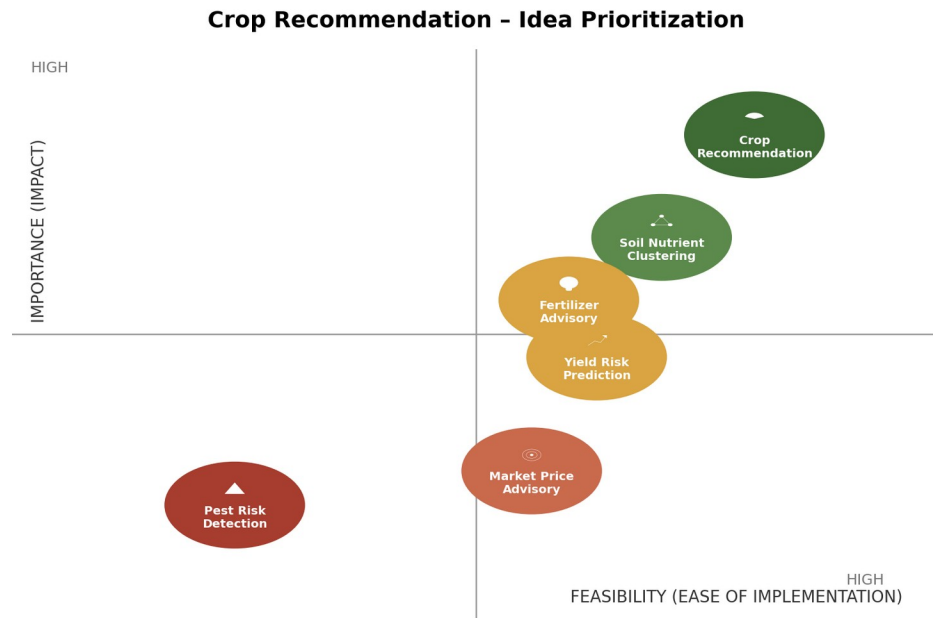
### Step-1: Selecting the Problem Statement:

*PROBLEM: How might we recommend the most suitable crop for a given soil and climate profile using Machine Learning?*

### Step-2: Idea Phase

| Idea A – Crop Advisory        | Idea B – Resource & Yield Optimization |
|-------------------------------|----------------------------------------|
| Crop Suitability Prediction   | Fertilizer Requirement Estimation      |
| Soil Nutrient Analysis        | AI-Based Soil Evaluation               |
| Automated Crop Advisory       | Crop Clustering Analysis               |
| Smart Farming Platform        | Water Requirement Prediction           |
| Real-Time Crop Recommendation | Seasonal Crop Planning                 |

## Step-3: Idea Prioritization



*Idea prioritization quadrant for OptiCrop*

## Ideation Phase

### Define the Problem Statements

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
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| Maximum Marks | 2 Marks                                                      |

### Customer Problem Statement Template:

|                                                                                                                |                               |                                                                                                      |
|----------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------|
|  <b>I AM</b>                  | Who are they?                 | Farmers, small and marginal landholders, agricultural advisors who need reliable crop guidance.      |
|  <b>I'M TRYING TO</b>         | What do they want to achieve? | Choose the crop that will grow best and yield the most on their land, right now.                     |
|  <b>BUT</b>                 | What's getting in their way?  | Crop selection is often based on habit or guesswork rather than actual soil and climate conditions.  |
|  <b>BECAUSE</b>             | Why is this happening?        | Manual soil-report interpretation, lack of real-time data analysis, and limited agronomic expertise. |
|  <b>WHICH MAKES ME FEEL</b> | How does this make them feel? | Uncertain and financially exposed to crop failure.                                                   |

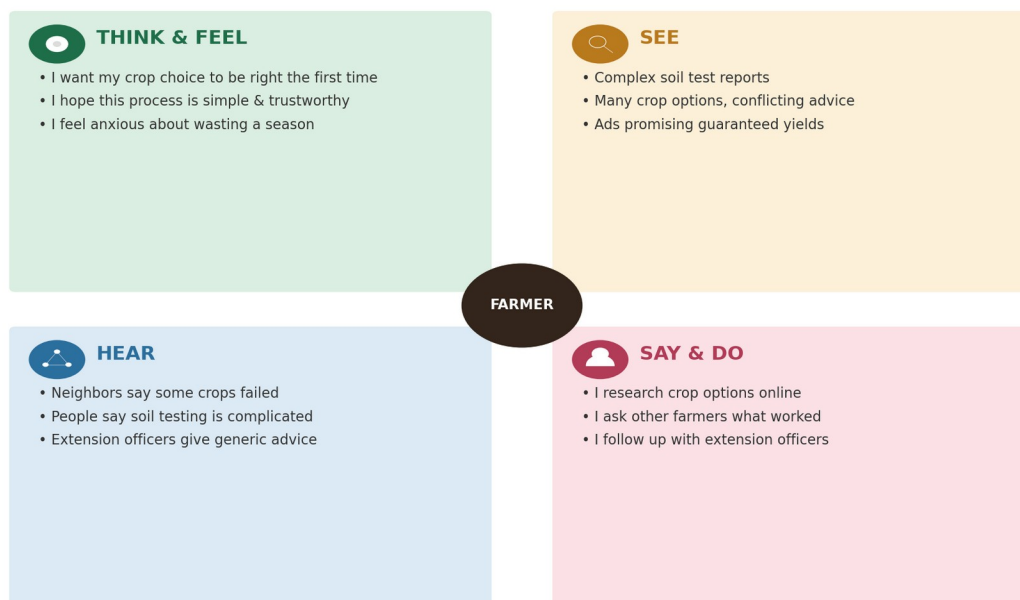
## Ideation Phase

### Empathize & Discover

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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### Empathy Map Canvas:

#### Empathy Map - OptiCrop Crop Recommendation System

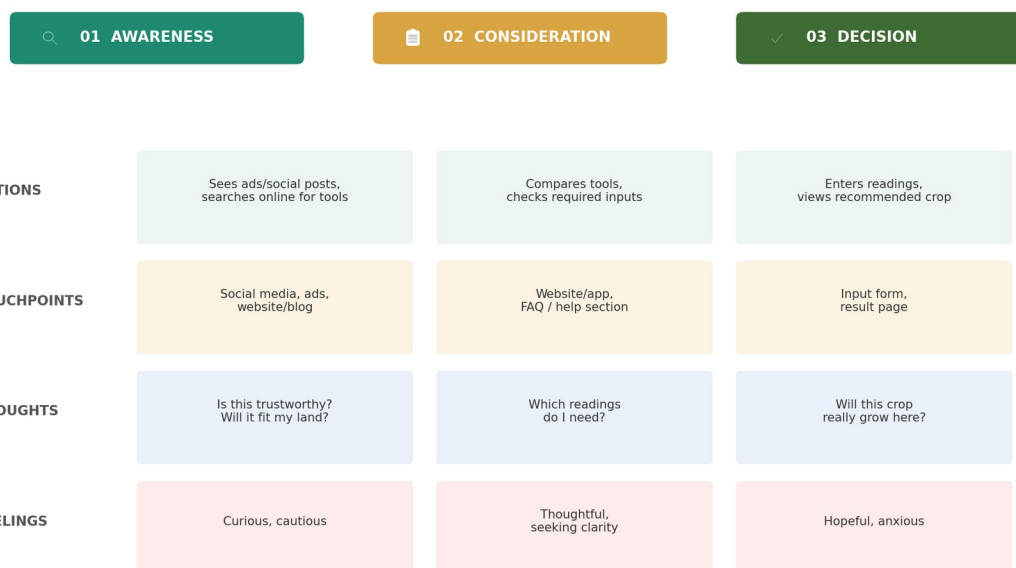




## Customer Journey Map

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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### Customer Journey Map - OptiCrop

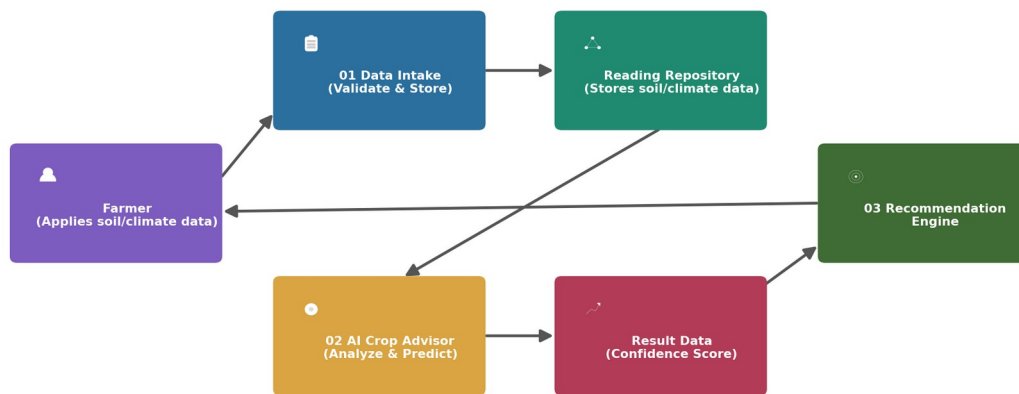


## Data Flow Diagram & User Stories

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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### Data Flow Diagrams:

OptiCrop - Smart Agricultural Production Optimization Engine



### User stories:

| User Type     | Epic                | US No. | User Story / Task                                                  | Priority | Release  |
|---------------|---------------------|--------|--------------------------------------------------------------------|----------|----------|
| Farmer        | Soil Data Entry     | USN-1  | As a user, I can enter my soil and climate readings.               | High     | Sprint-1 |
| Farmer        | Crop Recommendation | USN-2  | As a user, I can submit readings and get a crop recommendation.    | High     | Sprint-1 |
| Farmer        | View Crop Info      | USN-3  | As a user, I can view a short description of the recommended crop. | Medium   | Sprint-2 |
| Researcher    | Seasonal Insights   | USN-5  | As a researcher, I can view seasonal crop groupings.               | Medium   | Sprint-2 |
| Administrator | Model Monitoring    | USN-6  | As an administrator, I can monitor prediction accuracy.            | Medium   | Sprint-3 |





## Solution Requirements

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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### Functional Requirements:

| Functional Requirement | Sub Requirement                                               |
|------------------------|---------------------------------------------------------------|
| Soil & Climate Input   | Required fields: N, P, K, temperature, humidity, pH, rainfall |
| Crop Recommendation    | View the predicted crop                                       |
| Crop Information       | View a short description of the recommended crop              |
| Navigation             | View Home, About, and Find your crop pages                    |

### Non- Functional Requirements:

| Functional Requirement | Sub Requirement                                                                |
|------------------------|--------------------------------------------------------------------------------|
| Usability              | Application should provide a simple interface for all users.                   |
| Reliability            | Application should provide accurate and consistent crop recommendations.       |
| Performance            | Crop recommendation results should be generated within a few seconds.          |
| Availability           | Application should be available with minimal downtime.                         |
| Scalability            | Application should support multiple users and increasing requests efficiently. |

## Technology Stack

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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### Technical architecture:

The deliverable shall include the architectural diagram as below and the information.

| Components             | Description                                                                 | Technology                   |
|------------------------|-----------------------------------------------------------------------------|------------------------------|
| User interface         | Web interface for entering soil/climate data and viewing the recommendation | HTML, CSS                    |
| Application logic 1    | Page routing and navigation                                                 | Python (Flask)               |
| Application logic 2    | Soil/climate data validation and preprocessing                              | Python, Pandas               |
| Application logic 3    | AI-based crop recommendation                                                | Scikit-learn, NumPy          |
| Machine learning model | Crop recommendation model                                                   | Logistic Regression, K-Means |
| Infrastructure         | Application deployment on server or cloud                                   | Local server, Render         |

### Application characteristics:

| Characteristics          | Description                                 | Technology                     |
|--------------------------|---------------------------------------------|--------------------------------|
| Open source frameworks   | List the open-source frameworks used        | Flask, Pandas, scikit-learn    |
| Security implementations | Access controls to be used                  | HTTPS or local host            |
| Availability             | Justify the availability of the application | Local / cloud hosting          |
| Performance              | Design consideration for performance        | Pandas, NumPy, optimized model |

## Project Design Phase

### Solution architecture

| Field         | Value                                                        |
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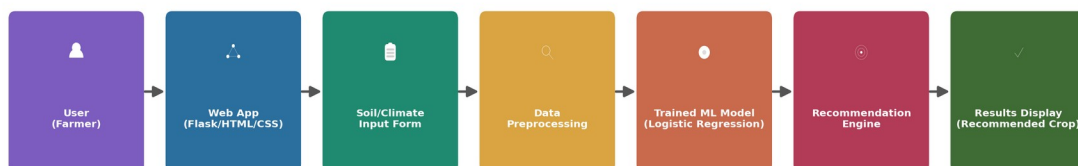
### Solution architecture:

Solution architecture is a process of designing a machine learning based web application that connects a farmer's soil and climate inputs with a crop recommendation.

It bridges the gap between crop selection challenges and an intelligent, data-driven software solution.

- It develops a reliable machine learning solution for recommending crops.
- It describes the system architecture, workflow, and functionality used to generate the crop recommendation.
- It provides a structured framework for developing, deploying, managing, and maintaining the crop recommendation system.

### Solution Architecture Diagram:



## Project design Phase

### Proposed solution

| Field         | Value                                                        |
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| Maximum Marks | 2 Marks                                                      |

### Proposed solution:

| Parameter                    | Description                                                                                                                             |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Problem statement            | Farmers often select crops based on habit rather than actual soil and climate suitability, reducing yield.                              |
| Idea or solution description | Develop a machine learning based web application that recommends the most suitable crop based on soil nutrients and climate conditions. |
| Novelty or uniqueness        | The system provides instant, data-driven crop recommendations backed by clustering and classification analysis.                         |
| Social impact                | Helps farmers make faster, more confident planting decisions and supports better use of resources.                                      |
| Revenue model                | The solution can be offered to agri-tech companies, cooperatives, and extension services as a software service.                         |
| Scalability of the solution  | The application can be scaled to support more crops, larger datasets, and additional regions.                                           |

## Project Design Phase

### Problem - solution fit

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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| Maximum Marks | 2 Marks                                                      |

### Project solution fit template:

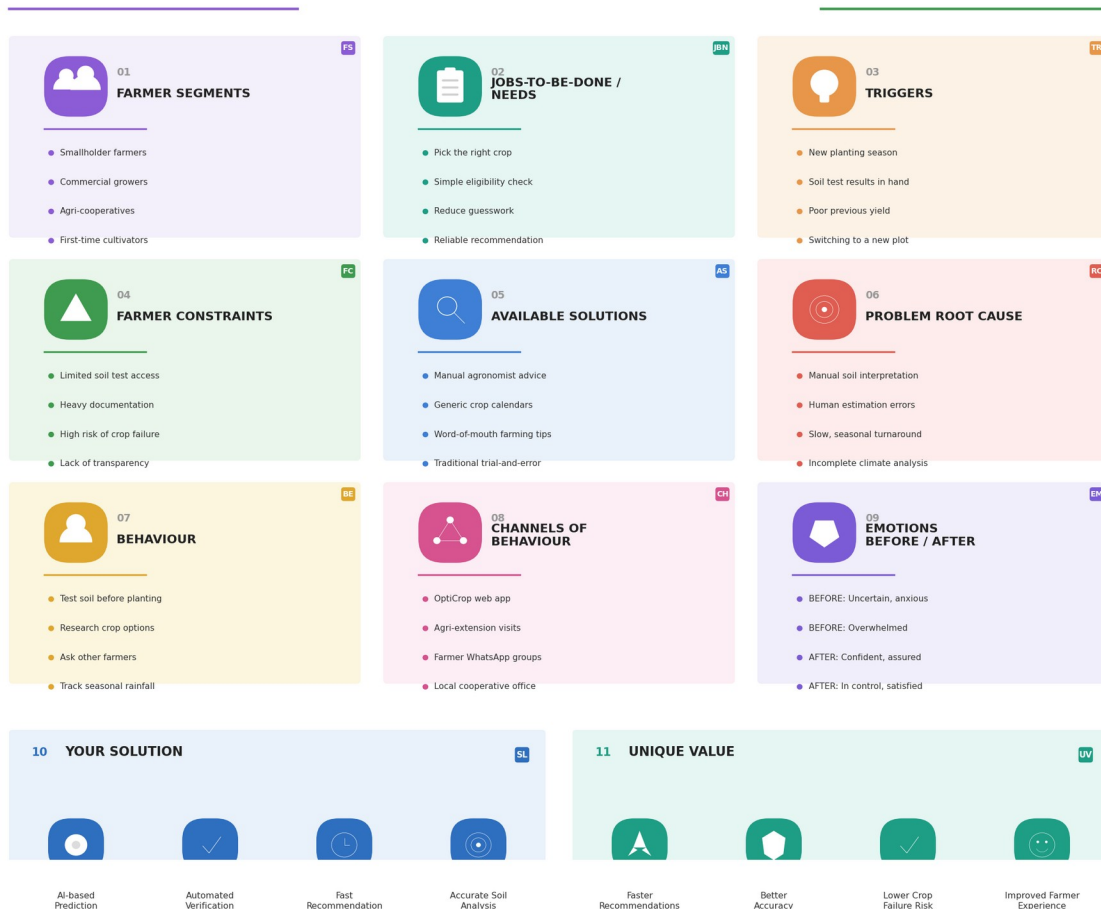
The problem-solution fit means identifying the challenges faced by farmers and agricultural stakeholders in selecting suitable crops, and providing a machine-learning-based solution that accurately recommends the best crop for given soil and climate conditions, reducing guesswork and improving yield outcomes.

### Main purpose:

- Reduces reliance on guesswork and improves crop selection accuracy.
- Improves farmer confidence with data-backed recommendations.
- Helps researchers and policymakers analyze crop-environment patterns.
- Enhances resource efficiency for water, fertilizer, and soil nutrients.
- Recommends the crop best suited to a specific plot's nutrient and climate profile.
- Helps farmers plan seasonal crop choices in advance.

## CROP RECOMMENDATION PROCESS & OPTIMIZATION FRAMEWORK

A holistic approach to streamline crop selection decisions



*Crop Recommendation Process & Optimization Framework*



## Project Planning Phase

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
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| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                      |

### Product Backlog, Sprint Schedule and Estimation:

| Sprint   | Epic                 | US No. | Task                                                        | Points | Priority |
|----------|----------------------|--------|-------------------------------------------------------------|--------|----------|
| Sprint-1 | Data Collection      | USN-1  | Collect the crop recommendation dataset for model training. | 2      | High     |
| Sprint-1 | Data Preprocessing   | USN-2  | Clean and preprocess the dataset.                           | 2      | High     |
| Sprint-2 | Clustering           | USN-3  | Apply K-Means clustering to identify soil/climate patterns. | 3      | Medium   |
| Sprint-2 | Model Training       | USN-4  | Train the Logistic Regression model for crop prediction.    | 3      | High     |
| Sprint-3 | Flask Integration    | USN-6  | Integrate the trained model with the Flask application.     | 3      | High     |
| Sprint-4 | Crop Recommendation  | USN-8  | Predict the most suitable crop from user input.             | 3      | High     |
| Sprint-4 | Testing & Deployment | USN-9  | Test and deploy the application successfully.               | 3      | Medium   |



## Code-Layout, Readability and Reusability

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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| Maximum Marks | 5 Marks                                                      |

This document evaluates the quality of the code base in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design.

| S.No | Code Quality Parameter    | Description                                   | Followed | Remarks                                                                             |
|------|---------------------------|-----------------------------------------------|----------|-------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes      | Python, HTML, CSS files follow proper indentation.                                  |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes      | Project is organized into data, model, app, templates, and static folders.          |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes      | Variables such as FEATURE_COLUMNS, CROP_INFO, and prediction are descriptive.       |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes      | Functions clearly indicate routing, prediction, and preprocessing.                  |
| 5    | Code Comments             | Inline and block comments explain logic       | Yes      | Important sections are documented to improve understanding and maintenance.         |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes      | Dataset generation, training, and the Flask app are separated into different files. |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes      | The implementation avoids unnecessary duplicate code and keeps modules concise.     |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes      | Basic validation and error handling are implemented to manage invalid user input.   |

### Reusable Components:

| S.No | Component / Module    | Technology Used | Purpose                   | Reusability |
|------|-----------------------|-----------------|---------------------------|-------------|
| 1    | Flask Web Application | Flask           | Handles user requests and | High        |





| S. No | Component / Module                    | Technology Used  | Purpose                        | Reusability |
|-------|---------------------------------------|------------------|--------------------------------|-------------|
|       | (app.py)                              |                  | recommendation workflow        |             |
| 2     | Crop Recommendation Model (model.pkl) | Scikit-learn     | Used for every crop prediction | High        |
| 3     | Data Preprocessing Module             | Pandas, NumPy    | Used before every prediction   | High        |
| 4     | HTML Templates                        | HTML5            | User Interface                 | Medium      |
| 5     | Static Resources                      | CSS3, JavaScript | UI styling and interaction     | Medium      |

### Overall Code Quality Assessment:

| Aspect                   | Rating (1–5) | Comments                                                                                                               |
|--------------------------|--------------|------------------------------------------------------------------------------------------------------------------------|
| Code Layout & Structure  | 5            | Well-organized project structure with separate folders for data, model, and app.                                       |
| Readability              | 5            | Code is easy to understand with meaningful names and proper formatting.                                                |
| Reusability              | 5            | Model, preprocessing logic, and Flask application can be reused or extended easily.                                    |
| Documentation / Comments | 5            | Code is documented sufficiently for understanding and future maintenance.                                              |
| Overall Score            | 5/5          | The project follows a modular, maintainable, and reusable architecture suitable for deployment and future enhancement. |



## Coding and solution

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
| Name          | K.Hemanth Kumar                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                      |

This section evaluates the quality and completeness of the implemented code and the overall solution, demonstrating working functionality, clean code practices, and alignment with the defined requirements.

### Solution Summary:

| Field                         | Details                                                                                                                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL         | <a href="https://github.com/&lt;your-username&gt;/OptiCrop">https://github.com/&lt;your-username&gt;/OptiCrop</a>                                                                                                                               |
| Programming Language(s)       | Python, HTML5, CSS3                                                                                                                                                                                                                             |
| Framework(s) Used             | Flask, scikit-learn                                                                                                                                                                                                                             |
| Key Features Implemented      | Crop recommendation using Machine Learning; data preprocessing; Flask-based web application; user-friendly soil/climate input form; real-time crop recommendation; trained model integration using pickle; input validation and result display. |
| Pending / Incomplete Features | Cloud deployment; database integration; user authentication; recommendation history storage.                                                                                                                                                    |
| Setup / Run Instructions      | 1. Clone the GitHub repository. 2. Install the required Python libraries using requirements.txt. 3. Open the project folder. 4. Run app.py. 5. Open the Flask application in the browser. 6. Enter soil/climate details and click Predict.      |

### Code Quality Checklist:

| S.No | Criteria                                              | Status |
|------|-------------------------------------------------------|--------|
| 1    | Code is modular and organized into functions/classes  | Yes    |
| 2    | Meaningful variable and function names are used       | Yes    |
| 3    | Code includes comments/documentation where necessary  | Yes    |
| 4    | Error handling is implemented for critical operations | Yes    |
| 5    | The application runs without critical errors          | Yes    |

## No. of Functional Features included in the Solution

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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| Maximum Marks | 5 Marks                                                      |

## Functional Features Overview

| Feature Name              | Feature Description                                                              | Module/<br>Component   | Status | Marks |
|---------------------------|----------------------------------------------------------------------------------|------------------------|--------|-------|
| Soil & Climate Input Form | Allows users to enter N, P, K, temperature, humidity, pH, and rainfall readings. | Frontend (HTML/CSS)    | Done   | High  |
| Input Validation          | Validates user input before processing.                                          | Flask Backend          | Done   | High  |
| Data Preprocessing        | Cleans and transforms input data into the required format.                       | Preprocessing Module   | Done   | High  |
| Crop Recommendation       | Predicts the most suitable crop using the trained ML model.                      | Machine Learning Model | Done   | High  |
| Result Display            | Displays the recommended crop and a short description to the user.               | Flask Application      | Done   | High  |
| Trained Model Loading     | Loads the saved machine learning model (model.pkl) for prediction.               | Model Loader           | Done   | High  |
| Responsive User Interface | Provides a user-friendly interface accessible through a web browser.             | HTML/CSS/JavaScript    | Done   | High  |
| Error Handling            | Handles invalid or incomplete user input gracefully.                             | Flask Backend          | Done   | High  |

## Performance testing

### Performance model test

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
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| Maximum Marks | 5 Marks                                                      |

### Test scenarios and results:

| Test ID | Scenario                   | Expected Result                                                    | Actual Result                                   | Pass/<br>Fail |
|---------|----------------------------|--------------------------------------------------------------------|-------------------------------------------------|---------------|
| FT-01   | User Input Validation      | Valid inputs accepted; invalid inputs show an error.               | Worked as expected.                             | Pass          |
| FT-02   | Crop Recommendation        | System predicts the most suitable crop.                            | Prediction displayed successfully.              | Pass          |
| FT-03   | Model Integration Check    | Application sends input to the ML model and receives a prediction. | Model returned a prediction successfully.       | Pass          |
| FT-04   | Application Navigation     | Pages load correctly without errors.                               | All pages loaded successfully.                  | Pass          |
| PT-01   | Response Time Test         | Prediction displayed within 3 seconds.                             | Prediction displayed within 3 seconds.          | Pass          |
| PT-02   | Multiple Prediction Test   | Application responds correctly without slowing down.               | Multiple predictions completed successfully.    | Pass          |
| PT-03   | Application Stability Test | Application should work without crashing.                          | Application remained stable throughout testing. | Pass          |



## Project Executable Files

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
| Name          | K.Hemanth Kumar                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 3 Marks                                                      |

| S.No | Item to Submit                               | Submitted |
|------|----------------------------------------------|-----------|
| 1    | Complete source code (all files and folders) | Yes       |
| 2    | README / Setup Guide                         | Yes       |
| 3    | requirements.txt                             | Yes       |
| 4    | Database schema / seed files (if applicable) | N/A       |
| 5    | Environment configuration (.env.example)     | No        |
| 6    | Deployed application URL                     | No        |
| 7    | APK / Executable binary                      | N/A       |
| 8    | Dockerfile / Containerization config         | No        |
| 9    | Test files and test results                  | Yes       |
| 10   | Demo video / walkthrough                     | Yes       |

### File / Folder Structure:

#### OPTICROP/

```

├── data/
│   ├── Crop_recommendation.csv    # Dataset used for training
│   └── generate_dataset.py        # Script that builds the dataset
├── model/
│   ├── train_model.py             # EDA, preprocessing, clustering, training
│   ├── model.pkl                 # Trained Logistic Regression model
│   └── plots/                    # EDA charts + classification report
├── app/
│   ├── app.py                    # Main Flask application
│   ├── static/style.css          # CSS styling
│   └── templates/                # base.html, home.html, about.html, findyourcrop.html

```

---

```
└─ requirements.txt      # Python dependencies
└─ README.md            # Project documentation
```

## Deployment / Access Details:

| Field                       | Value                                                                                                             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | Not Hosted (Runs Locally)                                                                                         |
| Login Credentials           | Not required – no authentication in this version                                                                  |
| Platform / Hosting Provider | Local Flask Server                                                                                                |
| Repository Link             | <a href="https://github.com/&lt;your-username&gt;/OptiCrop">https://github.com/&lt;your-username&gt;/OptiCrop</a> |
| Demo Video Link             | [project demo video link]                                                                                         |

## Run Instructions:

- Clone the repository.
- Install dependencies using requirements.txt.
- Run app.py.
- Open <http://127.0.0.1:5000> in your browser.
- Enter soil and climate details and click Predict.

## Known Issues / Limitations:

| S.No | Known Issue            | Workaround / Status            |
|------|------------------------|--------------------------------|
| 1    | Runs locally only      | Deploy to cloud                |
| 2    | No database            | Can integrate MySQL/PostgreSQL |
| 3    | No user authentication | Can be added later             |



## Communication

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
| Name          | K.Hemanth Kumar                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 1 Marks                                                      |

### Communication Plan:

| S. No | Type                      | Frequency | Channel / Tool               | Participants    | Purpose                                   |
|-------|---------------------------|-----------|------------------------------|-----------------|-------------------------------------------|
| 1     | Project Planning          | Daily     | Personal Notes, VS Code      | Self            | Plan daily development tasks              |
| 2     | Progress Review           | Weekly    | GitHub, VS Code              | Self            | Review completed modules and pending work |
| 3     | Issue / Bug Resolution    | As Needed | Stack Overflow, Docs, GitHub | Self            | Resolve coding and runtime issues         |
| 4     | Guide / Mentor Discussion | Weekly    | Google Meet / WhatsApp       | Student & Guide | Discuss progress and receive feedback     |
| 5     | Documentation Review      | Once      | Microsoft Word               | Self            | Verify documentation before submission    |
| 6     | Final Project Submission  | Once      | Skill Wallet & GitHub        | Self            | Submit final project and documentation    |

### Communication Challenges & Resolutions:

| S.No | Challenge Faced                                                         | Resolution / Action Taken                                                        |
|------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 1    | Difficulty integrating the trained crop recommendation model with Flask | Verified file paths, loaded model.pkl correctly, tested predictions locally.     |
| 2    | Errors due to missing Python libraries                                  | Installed required packages using pip install -r requirements.txt.               |
| 3    | Preparing documentation according to the SmartBridge template           | Updated every section with OptiCrop-specific content, diagrams, and screenshots. |



## Demonstration of proposed features

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
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| Maximum Marks | 1 Marks                                                      |

### Feature Implementation Status:

| S.No | Feature Name                              | Status      | Demonstrated | Remarks                              |
|------|-------------------------------------------|-------------|--------------|--------------------------------------|
| 1    | Crop Recommendation                       | Implemented | Yes          | Prediction generated successfully    |
| 2    | Soil & Climate Input Form                 | Implemented | Yes          | User-friendly interface              |
| 3    | Machine Learning Model Integration        | Implemented | Yes          | Model loaded successfully            |
| 4    | Responsive Web Interface                  | Implemented | Yes          | Supports desktop and mobile browsers |
| 5    | Result Display                            | Implemented | Yes          | Results displayed correctly          |
| 6    | Project Documentation & GitHub Repository | Implemented | Yes          | Repository submitted successfully    |

### Feature Implementation Summary:

| Metric                          | Value |
|---------------------------------|-------|
| Total Features Proposed         | 6     |
| Total Features Implemented      | 6     |
| Total Features Demonstrated     | 6     |
| Overall Implementation Rate (%) | 100%  |



## Project Demo Planning

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
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| Maximum Marks | 1 Marks                                                      |

| S. No | Demo Section                                                       | Duration | Responsible |
|-------|--------------------------------------------------------------------|----------|-------------|
| 1     | Introduction – objectives, problem statement, expected outcomes    | 2 min    | Self        |
| 2     | Solution Overview – architecture, workflow, dataset, ML model      | 2 min    | Self        |
| 3     | Application Demonstration – live crop recommendation               | 3 min    | Self        |
| 4     | Feature Explanation – validation, preprocessing, model integration | 3 min    | Self        |
| 5     | GitHub Repository & Documentation walkthrough                      | 2 min    | Self        |
| 6     | Conclusion & Future Enhancements                                   | 2 min    | Self        |



## Scalability & Future Plan

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
| Name          | K.Hemanth Kumar                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 1 Marks                                                      |

### Current System Limitations:

| S. No | Limitation                           | Impact                                   | Priority |
|-------|--------------------------------------|------------------------------------------|----------|
| 1     | No user authentication               | Unauthorized access may occur            | High     |
| 2     | No database integration              | Recommendation history not stored        | Medium   |
| 3     | Limited deployment (local Flask app) | Restricted scalability and accessibility | High     |

### Future Roadmap:

| Phase   | Planned Feature                                | Timeline   | Expected Impact                            |
|---------|------------------------------------------------|------------|--------------------------------------------|
| Phase 1 | Cloud deployment and database integration      | 1–2 Months | Improved accessibility and data management |
| Phase 2 | User authentication and admin dashboard        | 2–3 Months | Enhanced security and administration       |
| Phase 3 | Recommendation history and analytics dashboard | 3–4 Months | Better monitoring and reporting            |
| Phase 4 | Advanced ML models and API integration         | 4–6 Months | Higher accuracy and enterprise scalability |

## Team Involvement in Demonstration

| Field         | Value                                                        |
|---------------|--------------------------------------------------------------|
| Date          | 7/9/2026                                                     |
| Name          | K.Hemanth Kumar                                              |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 1 Marks                                                      |

| Team Member            | Role                  | Contribution                                                                                                                                | Participation |
|------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Self (K.Hemanth Kumar) | Presenter & Developer | Developed the project, demonstrated the application, explained the machine learning model, answered questions, and presented documentation. | Active        |

| Aspect                                 | Details                      |
|----------------------------------------|------------------------------|
| Team Leader / Coordinator              | Self (K.Hemanth Kumar)       |
| Overall Team Coordination Rating (1-5) | 5/5                          |
| Any issues during demo                 | No major issues encountered. |



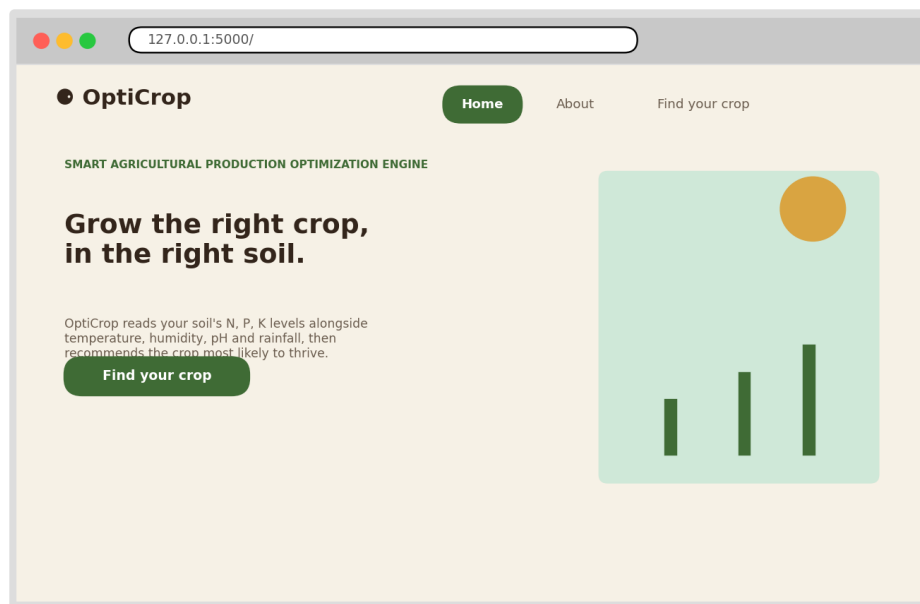
## Output Screenshots

OptiCrop's crop recommendation system was successfully developed and tested using a Flask-based web application. The application accepts user input such as Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall. Based on the entered details, the trained machine learning model recommends the crop most likely to succeed.

*Note: the previews below are stylized page mockups matching the app's actual design tokens. Swap in real browser screenshots after running the app locally if your submission requires exact captures.*

### Home page

The home page of the OptiCrop application provides an overview of the crop recommendation system. It introduces the purpose of the application and allows users to navigate to the crop recommendation form by clicking the “Find your crop” button.



### Find your crop – input form

The form allows users to enter Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall values. These details are used by the machine learning model to recommend the most suitable crop.



127.0.0.1:5000/findyourcrop

OptiCrop Home About Find your crop

CROP RECOMMENDATION

**Find your crop**

Nitrogen (N)

Phosphorous (P)

Potassium (K)

Temperature

Humidity

pH

Rainfall

Recommend a crop

Your result will appear here

## Crop recommendation result

After submitting valid soil and climate details, the machine learning model analyzes the input and recommends the crop most likely to succeed, along with a short description. This confirms that the application performs real-time crop recommendation successfully.

127.0.0.1:5000/findyourcrop

OptiCrop Home About Find your crop

(soil & climate form filled in)

RECOMMENDED CROP

**rice**

Thrives in warm, humid conditions with heavy rainfall.

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## Conclusion

OptiCrop – Smart Agricultural Production Optimization Engine demonstrates how Machine Learning can be effectively applied to improve crop selection decisions in agriculture. By analyzing soil nutrients and climate conditions – Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall – the system accurately recommends the crop best suited to a given plot, enabling faster and more confident planting decisions.

The integration of the trained Logistic Regression model with a Flask-based web application provides a simple, interactive, and user-friendly platform for real-time crop recommendation. The system also groups agricultural conditions using K-Means clustering, offering additional insight into crop-environment relationships.

Overall, the project reduces reliance on habit-based crop selection, minimizes guesswork, improves resource efficiency, and supports farmers and agricultural stakeholders in making data-driven planting decisions. It highlights the practical use of Artificial Intelligence and Machine Learning in solving real-world agricultural problems while enhancing productivity and sustainability. OptiCrop serves as a scalable foundation that can be further enhanced with advanced Machine Learning models, larger and region-specific datasets, cloud deployment, and stronger data management to meet the evolving needs of modern agriculture.